Thinking Without Knowing: On Large Language Models and the Simulation of Mind

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Abstract

Large Language Models (LLMs) such as GPT-style transformers have reignited debates on the nature of thinking and whether these systems *genuinely* think or merely simulate thought. Blending technical and philosophical analysis, this paper assesses LLMs against criteria drawn from functionalism, the Chinese-Room argument, Chalmers' "easy/hard" distinction, and the symbol-grounding problem. We update the discussion with recent work that frames LLMs as *mimics, not minds* and foreground the often neglected question of *agency*: current models pursue no intrinsic goals and act only when prompted. After outlining transformer architectures, pre-training, and in-context learning, we examine originality, intentionality, grounding, and autonomy to ask whether a convincing simulation ever suffices for real thought. We conclude that today's LLMs powerfully *simulate* aspects of mind while lacking the conscious awareness, intrinsic intentionality, and self-directed agency that would justify calling them full-fledged thinkers.

Keywords: Large Language Models; Philosophy of Mind; Intentionality; Agency; Machine Consciousness.

1 Introduction

Can a machine with no consciousness or understanding nonetheless *think*? This question, famously posed and reformulated by Alan Turing in 1950, remains central to artificial intelligence research[1]. Turing sidestepped the vague terms in the question "Can machines think?" by proposing an operational test of indistinguishability in conversation - the Imitation Game - as a proxy[1]. Today, the advent of large language models (LLMs) that can hold coherent conversations and solve problems brings renewed urgency to this question. Models like GPT-3 and its successors produce human-like responses across countless topics, prompting both excitement and philosophical unease. They appear to engage in

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reasoning, answer complex questions, and even generate creative prose, all without any conscious awareness or experiential world grounding. This paradox of *thinking without knowing*, wherein an entity might exhibit intelligent behaviour with no genuine understanding, challenges our definitions of mind and thought.[1]

This paper asks whether LLMs should count as *thinkers* and, if not, what exactly is missing. Beyond the classical criteria of functional[2] equivalence and semantic grounding, we add two dimensions highlighted by recent authors: (i) *autonomy and agency* - whether a system pursues its own goals (Godfrey-Smith [17]) - and (ii) *depth of understanding versus mere simulation* (Smith [12], Bender–Koller [14]). We therefore proceed in two strands. First, we frame the problem philosophically: What does it mean to think, and what criteria must an entity meet to qualify? We review functionalism - the view that mental states are defined by functional roles, not by their material substrate - as articulated by Putnam and Fodor[2, 3]. We then consider Searle's Chinese Room thought experiment, which argues that symbol manipulation alone (the basis of computer programs) is insufficient for understanding[4]. We introduce Chalmers' distinction between the "easy" problems of explaining cognitive functions and the "hard" problem of explaining subjective experience[5]. We also outline Harnad's symbol grounding problem, the challenge of how symbols in a purely formal system acquire meaning tied to the real world[6]. Second, we provide a technical overview of LLMs: how transformer-based architectures work and learn from data, and what capabilities they exhibit. By blending these perspectives, we aim to clarify the key issues of originality, intentionality, grounding of meaning, and the relationship between simulation and reality of mind. We then present arguments on both sides - those who see LLMs as genuine (if unconventional) thinkers and those who remain skeptical. Finally, we offer a nuanced conclusion on whether these models genuinely think or merely simulate thought, and whether that distinction makes a practical difference.

2 Background

2.1 Philosophical Perspectives on Mind and Thinking

One influential perspective in the philosophy of mind is **functionalism**. Hilary Putnam and Jerry Fodor were early proponents of this view, which defines mental states by their functional or causal roles rather than by their physical composition[2, 3]. In Putnam's words, the mind is characterised by what it *does*, not what it is made of: "We could be made of Swiss cheese and it wouldn't matter" so long as the functional organisation that gives rise to mental processes is preserved[2]. This idea implies a thesis of *multiple realisability*: a given mental state (say, pain or belief) could be implemented in various physical substrates

[1] For contrasting evaluations see Schwitzgebel [10] on LLMs as "mimics, not minds" and Millière [11] on their grounding deficit.

[2] Functionalism: the view that mental states are defined by causal/functional roles rather than substrate.

(biological neurons, silicon circuits, etc.) as long as the causal relationships between inputs, internal states, and outputs are the same[2][3]. Functionalism thus opens the door for AI systems to have minds - if the functional structure of their operations mirrors that of thinking beings, the *medium* (biological or electronic) is irrelevant to whether they have mental states.

John Searle's **Chinese Room** argument presents a critical counterpoint to functionalism and to what he calls "strong AI." In a 1980 thought experiment, Searle imagines himself (an English speaker with no knowledge of Chinese) in a room following a set of rules (a program) to manipulate Chinese symbols[4]. To outsiders, the outputs from the room are indistinguishable from a fluent Chinese speaker's answers, yet Searle himself understands nothing of the Chinese content. According to Searle, this scenario illustrates that merely executing formal rules (syntactic operations on symbols) is not sufficient for *semantics* or understanding[4]. The person in the room (or by analogy, a computer running a program) has no grasp of meaning or intentionality; they are, in Searle's terms, simply manipulating symbols based on shape, "following syntactic rules without semantic comprehension"[4]. Searle thus concludes that no matter how intelligent a machine's behaviour seems, the machine (like the man in the room) would not *actually* understand anything; it would merely simulate understanding. In formal terms, Searle's argument rests on two claims: (1) brains cause minds, and (2) "syntax doesn't suffice for semantics"[4]. He distinguishes **strong AI** (the claim that an appropriately programmed computer *literally has* a mind and understanding) from **weak AI** (the view that computers only simulate thought, producing as-if intelligence that can be useful for research, but not real understanding)[4]. The Chinese Room thought experiment targets strong AI by asserting that running the right program (no matter how human-like the responses) would not confer a mind or genuine understanding[4]. This poses a direct challenge to functionalism and computational theories of mind that equate simulating a mental process with instantiating one[4].

Another crucial concept is **intentionality**, the "aboutness" of mental states - the way thoughts and beliefs can be *about* things in the world. Philosophers like Franz Brentano and John Searle have argued that genuine minds have *intrinsic* intentionality, whereas artifacts have only *derived* intentionality (meaning assigned by their designers or users)[4][3]. Searle contends that human mental states intrinsically refer to objects and states of affairs, arising from conscious experience, whereas a machine's states only have meaning insofar as we interpret them as such[4]. For example, a thermostat might be said to "want" to keep the room at 20°C, but this is merely us ascribing purpose to it; the thermostat itself has no concept of *wanting* or any awareness of temperature - it simply follows its programmed response. By this view, an LLM does not truly *intend* to inform or deceive or answer; it has no goals or understanding of the content, only an implicit statistical mapping from prompts to responses. All the "aboutness" in its outputs (say, talk about world events or factual claims) is parasitic on the

[3]Original intentionality (Searle 1980) means a mental state intrinsically 'about' an object; derived intentionality is ascribed by observers to mere symbols.

meanings humans imbued in the training text and in our interpretation of its replies. This gap in intentionality is a major point for skeptics who argue LLMs lack any genuine mental states.

Chalmers' distinction between **"easy" and "hard" problems of consciousness** adds another layer to the analysis. The "easy" problems (so called tongue-in-cheek by Chalmers) are those of explaining cognitive functions and behaviours: how do we discriminate stimuli, integrate information, produce verbal reports, etc. These are considered "easy" not because they are trivial, but because they seem amenable to standard scientific explanation in terms of neural or computational mechanisms[5]. In contrast, the "hard problem" is explaining *subjective experience* or qualia - why and how it *feels* like something to have a mind[4][5]. One can imagine a hypothetical being (often called a *philosophical zombie*) that behaves exactly like a human, solving the easy problems, but has no conscious experience. Many philosophers (Chalmers included) consider that an intuitive possibility, highlighting an explanatory gap in any purely functional account of mind[5]. In the context of AI, an LLM might conceivably replicate many intelligent behaviours (the "easy" problems of language understanding, question answering, etc.) without any inner life or consciousness. If consciousness or subjective awareness is deemed an essential component of *thinking*, then LLMs at best solve the easy problems but not the hard problem - they might be seen as "zombies" that talk like a person but experience nothing. Not all scholars agree the hard problem is a real issue; Type-A materialists such as Daniel Dennett and others argue that once we've explained all the cognitive functions, there is no additional mystery and that consciousness is not something beyond those functions[7]. This disagreement is important: if one believes only functional, behavioural criteria matter (a stance consistent with Dennett or with functionalism), then an AI that passes those criteria could be considered as thinking. But if one believes that without subjective consciousness there is no genuine thinking (a more dualist or at least non-reductionist view), then LLMs clearly fall short.

Finally, the **symbol grounding problem** identified by Stevan Harnad highlights the challenge of how symbols gain meaning. In 1990, Harnad asked: how can the "semantic interpretation of a formal symbol system" be made intrinsic to the system, rather than just "parasitic on the meanings in our heads"[6]? In other words, when a computer manipulates symbols (words, tokens, etc.), those symbols mean nothing to the machine unless they are grounded in something the machine can relate to its own experience or environment. Humans learn the meanings of words ultimately through perception and interaction (we ground the word "apple" in our sensorimotor experience with actual apples, for instance). An AI that only processes text might never connect its symbols to actual objects or experiences - it lives in a world of "meaningless symbols" that only have meaning when interpreted by humans who supply the semantics[6]. Harnad equates the symbol grounding problem with the problem of intrinsic meaning or intentionality noted by Searle[4]. For LLMs, this issue is acute: these models learn purely from text. Does a word like "apple" in an LLM's memory correspond to anything like what we humans mean by apple (the round fruit, with certain

taste and texture)? Or is it just a pattern of co-occurrence with other words (like "red," "tree," "pie") with no direct sensory referent? The symbol grounding problem suggests that without some grounding, an AI's "understanding" of words is fundamentally different from human understanding. It might explain why LLMs sometimes make absurd mistakes: without ground truth from reality, they have only statistical correlations to go on. In summary, philosophy of mind provides several lenses - functional role, intentionality, consciousness, and grounding - through which to assess whether an AI is truly thinking or merely appearing to do so.

2.2 Large Language Models: Architecture and Capabilities

LLMs like GPT-3 (and its sucessors) are built on modern **transformer** architectures. The transformer is a deep neural network design introduced by Vaswani et al. in 2017[8]. Unlike earlier recurrent neural networks, transformers use a mechanism called *self-attention* to process input sequences. In a transformer, the text is first broken into tokens (subword units), which are mapped to vector embeddings[8]. Then, in each layer of the network, tokens exchange information with all other tokens via multi-head self-attention: the model computes pairwise attention scores to determine which other words in the context are most relevant to each word[8]. This allows the model to capture long-range dependencies and the context of a word from the entire sequence, rather than just neighboring words. Multiple layers of self-attention and feed-forward transformations are stacked to build a powerful encoder and/or decoder that can model complex patterns in text. Transformers dispense with the need for recurrent loops; they process sequences in parallel, which makes training on very large datasets more efficient[8]. The architecture is typically composed of an encoder (to read input text) and a decoder (to produce output text), though purely decoder-based transformers are often used for generative language modeling.

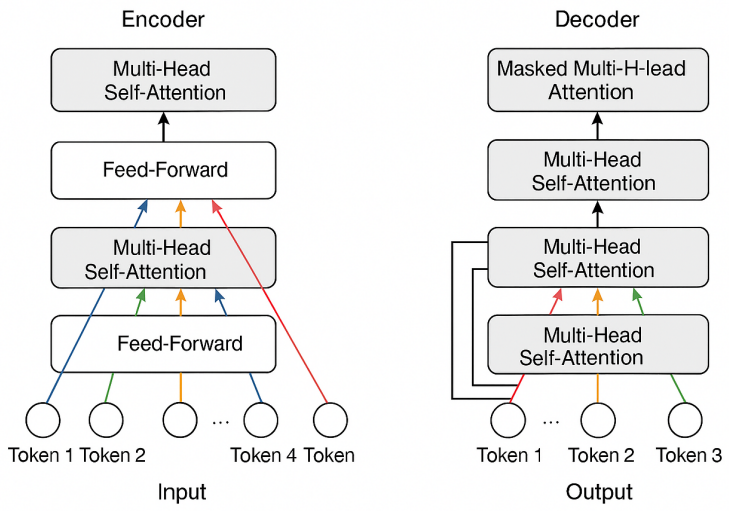

Figure 1: Schematic of a Transformer Architecture, with an encoder (left) and decode (right) [8].

The transformer design enabled training of much larger models on much larger datasets than previously possible. Subsequent innovations and scaling led to the development of **large language models** - transformer networks with billions of parameters (connection weights) trained on enormous text corpora. These models are typically trained with a simple objective: given a sequence of text, predict the next word (or token). Through this *self-supervised learning* on vast amounts of text (for example, internet scrapes, libraries of books, Wikipedia, etc.), LLMs learn statistical patterns of language at many levels[8]. They learn word meanings (as contextualised by usage), syntactic structures, facts and common sense associations, even styles and idioms - all implicitly encoded in the model's parameters from seeing many examples. A key point is that this training is unsupervised in the sense that no explicit human labeling is required; the model learns from raw text by trying to predict parts of it, gradually improving its internal representation of language.

After the broad **pretraining** phase on general text, an LLM can be adapted to specific uses through **fine-tuning**. In fine-tuning, the pretrained model is further trained on a narrower dataset or with additional objectives to specialise it or align it with desired behaviours[8]. For instance, an LLM can be fine-tuned on dialogue transcripts to better perform as a conversational agent, or fine-tuned on code to better generate programming solutions. Recent practice also includes fine-tuning with human feedback (reinforcement learning from human feedback, RLHF) to make the model's outputs more helpful and aligned with user intentions. The result of these processes is a model that possesses a broad, general capacity to generate and continue text in ways that often appear

remarkably intelligent.

LLMs exhibit several notable **capabilities**. First, they are *generative*: they can produce extended passages of coherent text, ranging from answering questions and summarising articles to writing stories or code. They do so by probabilistically sampling one word at a time, in a way that statistically fits the input prompt and the training distribution of language. Second, they demonstrate **in-context learning** abilities. In-context learning means that an LLM can adapt to a new task or follow instructions purely from the text prompt, without any further parameter updates or training[8][9]. For example, if you give an LLM a few examples of a task (say, translation from English to French) in the prompt and then a new English sentence, the model will often be able to produce the French translation. It is as if the model "learns" from the examples on the fly, performing what is also called **few-shot learning**[8, 9][4]. The surprising effectiveness of this was highlighted in the GPT-3 paper, titled "Language Models are Few-Shot Learners," which showed that a sufficiently large model (175 billion parameters) could achieve decent performance on many tasks with only a few demonstrations or even just a clear instruction, without gradient descent or weight updates[9]. This emergent capability suggests that LLMs can implicitly encode many tasks and can be guided to execute them by context alone.

Despite their lack of explicit world knowledge beyond text, LLMs also display a wide range of **world knowledge and reasoning**. They can answer trivia questions, explain jokes, write essays about abstract concepts, and so on. Much of this comes from pattern recognition over text data; for example, if the model has seen many instances of cause-and-effect described, it can mimic that reasoning pattern in new situations. Some reasoning tasks, especially those requiring multiple steps, can be challenging for LLMs, but prompt techniques (like asking the model to chain its reasoning steps explicitly) can help, indicating that the capacity for reasoning is latent in these models and can be activated with the right prompt.

However, LLMs also have well-documented **limitations**. They can produce factually incorrect statements (so-called "hallucinations") because they have no direct reference check on reality - only what text suggests is plausible. They may also generate internally inconsistent or nonsensical answers, especially if prompted in ambiguous or adversarial ways. They have no persistent memory of interaction (beyond the given context window), no genuine understanding of time, and no intrinsic goals or motivations. These limitations are often the flip side of their design: they predict text based on patterns, without any actual comprehension of what the text means or any model of the world beyond those patterns. In the next section, we delve into the heart of the debate: given what LLMs are and do, in what sense might they be said to "think"? What do these impressive capabilities amount to philosophically?

[4]Few-shot learning refers to an LLM's ability to adapt to a new task when given only a handful of annotated examples, without re-training its parameters.

3 Related Work on LLMs and the Philosophy of Mind

A rapidly growing literature examines whether large language models merely *simulate* mental processes or instantiate them. Schwitzgebel argues that present-day LLMs are best described as "mimics, not minds," insisting that genuine mentality requires more than the production of mind-like outputs[10]. Building on the classic symbol-grounding debate, Milliere updates Harnad's problem for text-only systems, contending that purely linguistic training deprives LLMs of any route to intrinsic meaning[11]. B C Smith echoes this worry in characterising LLMs as "glorified pattern matchers" that operate at the level of form rather than semantics [12] . Together these works frame the central dilemma: impressive linguistic performance forces a re-examination of functionalist criteria, yet lack of grounding and understanding motivates continued scepticism.

4 Main Argument: Do LLMs Think, or Only Simulate Thinking?

Equipped with the philosophical frameworks and an understanding of LLMs' workings, we can now ask: Does an LLM *think*? To answer, we must clarify what we count as thinking. If thinking is defined behaviourally or functionally (as in Turing's and the functionalists' approach), then the case for LLMs as thinkers is stronger. If thinking is defined by having conscious understanding or intentional mental states, the case is much weaker. In this section, we consider several key criteria and characteristics often associated with thinking, and evaluate how LLMs measure up:

- **Originality and Creativity:** Human thinking is often creative and generative of new ideas. Do LLMs produce anything genuinely original, or are they merely remixing and regurgitating existing text? On one hand, LLM outputs are derived from their training data; they cannot truly *invent* new knowledge beyond what patterns of words suggest. Critics liken them to "stochastic parrots," stitching together familiar phrases without true innovation[13]. The very term *stochastic parrot* implies that an LLM is randomly echoing the data it was fed, without understanding[13]. Indeed, LLMs have been shown to sometimes plagiarise or closely copy parts of their training data when prompted in certain ways. On the other hand, the space of possible combinations of words is astronomically large, and LLMs often generate sentences or stories that are not verbatim from any source. In fact, by recombining concepts in novel ways, they can produce outputs that humans might consider creative (for example, writing a poem on a whimsical prompt or proposing an unusual solution to a problem). One might argue that much human creativity also works by remixing existing ideas in new configurations. If an LLM produces a coherent, contextually appropriate piece of text that wasn't explicitly in

its training set, is that not a form of originality? The difference is that human creativity is usually linked to an intentional process - an author consciously aiming to create meaning - whereas an LLM's "creativity" is an accidental byproduct of statistical patterns. There is no intentional agent in the machine that *wants* to express something new. So, while LLMs can simulate creativity and occasionally surprise us with novel outputs, whether this counts as genuine originality is debatable. It may hinge on whether one believes creativity requires intentional thought or whether generating novel combinations is sufficient.

- **Understanding and Intentionality:** We typically assume that thinking involves *understanding* the content of one's thoughts. When a person thinks about "Paris," their mental state is *about* the city of Paris; it has intentionality. Do LLMs understand what they talk about? The consensus among researchers is that current LLMs lack such understanding - they have no grounded model of the world to which their words refer. B. C. Smith characterises them as "glorified pattern matchers" that operate on syntactic form without accessing semantic content [12]. Concretely, they treat words as abstract tokens, statistical entities whose relations are learned from text. When an LLM answers a question about Paris, it does not conjure a mental image of the Eiffel Tower or retrieve a fact it *knows* in a reflective sense; it simply emits words that frequently co-occur in discussions of Paris ("capital," "France," "Eiffel Tower"). Thus its output can be correct and informative, yet it would be misleading to say the model *knows* or believes those facts.

  Searle would say the LLM has only *syntax*, not *semantics*; it manipulates symbols without grasping their meaning [4]. Moreover, an LLM has no *intentions* or goals behind its words - it holds no desires or beliefs, and gains nothing from the conversation. It generates whatever is probabilistically appropriate to the prompt. In philosophy, this is the distinction between *original intentionality* (possessed by minds) and *derived intentionality* (possessed by words or symbols humans interpret) [2]. The content of LLM-generated text has meaning to *us*, the readers, because we map it onto our knowledge of the world; to the model, there is no "aboutness" - its internal states do not point to Paris or anything outside the text context. Proponents might respond that the model's latent representations *do* correlate with real-world concepts (activation patterns for "Paris" differ from those for "London"), and could claim that constitutes a shallow kind of understanding. Yet without experiential familiarity or semantic grounding, such "understanding" remains derivative. Unless one adopts an extreme behaviourist stance (understanding is as understanding does), LLMs fall short of genuine intentionality.

- **Grounding in Reality:** Following from the above, a thinker typically has some connection to the world - our thoughts ultimately link to perceptions and actions (even abstract thoughts build on metaphorical or experiential

foundations, according to some cognitive science theories). LLMs lack any embodiment or sensorimotor experience. They exist as disembodied text processors. This absence leads to Harnad's symbol grounding problem: their symbols (words) are not grounded in anything but other symbols[6]. For example, a child learning the word "apple" will eventually connect it to the taste, feel, and appearance of real apples. An LLM learns "apple" purely from how it is used in sentences ("apple pie," "red apple," "Apple Inc."). It develops a complex web of associations for the word, but those associations ultimately trace back to more words, not to physical reality. Without grounding, can we say the model *knows* what an apple is? This ungrounded nature is evident when LLMs make mistakes that no human with actual experience would, such as claiming absurd falsehoods with confidence. They lack the feedback loop that grounded agents have from the world to correct their internal representations. Now, some arguments suggest that with enough textual data, an LLM can acquire *implicit* grounding - text contains a lot of information about the world, after all. For instance, an LLM can learn that "fire is hot" and "fire can burn you" from reading, even if it never felt heat. Does that count as a form of grounding, albeit second-hand? It's indirect and brittle; the model might know the words "fire" and "hot" go together, but it won't flinch from a flame. Thus, most would maintain that without direct interaction with the world, the LLM's "knowledge" remains shallow and ungrounded. Some AI researchers are exploring ways to ground LLMs by integrating them with vision or robotics, but in their current form as pure text models, their world model is entirely derived from linguistic patterns.

- **Simulation vs. Instantiation of Mind:** Perhaps the most profound question is whether simulating the outward behaviour of thinking is equivalent to actually thinking. This echoes a core debate from Searle's Chinese Room and related discussions[4]. A strong functionalist position is that if a system functions indistinguishably from a thinking entity, then it *is* a thinking entity. In this view, the distinction between simulation and reality of mind is a false dichotomy: minds *are* what minds *do*. If an LLM can engage in dialogue, answer questions, reason through problems (at least apparently), then it is implementing the cognitive functions we associate with thinking. We might say it *has* those mental states in virtue of the functional role they play. This is essentially taking the Turing Test criterion seriously - if you cannot tell the difference between the AI and a human in conversation, then attributing thought to the AI is justified. Some philosophers and cognitive scientists (and science fiction writers) even speculate that a sufficiently advanced AI might develop self-awareness or other mind-like qualities simply by virtue of carrying out the same complex information-processing as a human brain. However, the opposing view, articulated by Searle and others, insists that simulation is not instantiation: no matter how perfect the performance, if the internal mechanism is just symbol manipulation without intentionality or

experience, then it is not actually thinking[4]. Searle famously analogised: "simulating X is not the same as having X." You can simulate digestion with a machine that churns and gurgles, but that doesn't mean the machine is actually digesting anything[4]. Likewise, an LLM might simulate the verbal output of a thinking mind, but have no mind. This perspective holds that there is a fact of the matter about being a thinker that goes beyond behaviour; something akin to an inner life or at least intrinsic semantic content is required. LLMs, according to this view, are more like wind-up dolls reciting impressive lines - a clever illusion rather than genuine cognition.

Understanding vs. Simulation: A core question is whether fluent linguistic behaviour suffices for *understanding*. Bender and Koller argue that current LLMs achieve "form without meaning," capturing surface regularities while lacking semantic competence[14]. Crane sharpens the point: genuine understanding, he claims, presupposes conscious or perceptual acquaintance with the world - an acquaintance absent in text-only models[15]. Our analysis supports their verdict: LLMs compute statistically apt continuations but do not *grasp* the propositional content they manipulate. Hence their success illustrates the power - and the limits - of simulation; they replicate the outputs of understanding without instantiating the underlying cognitive state.

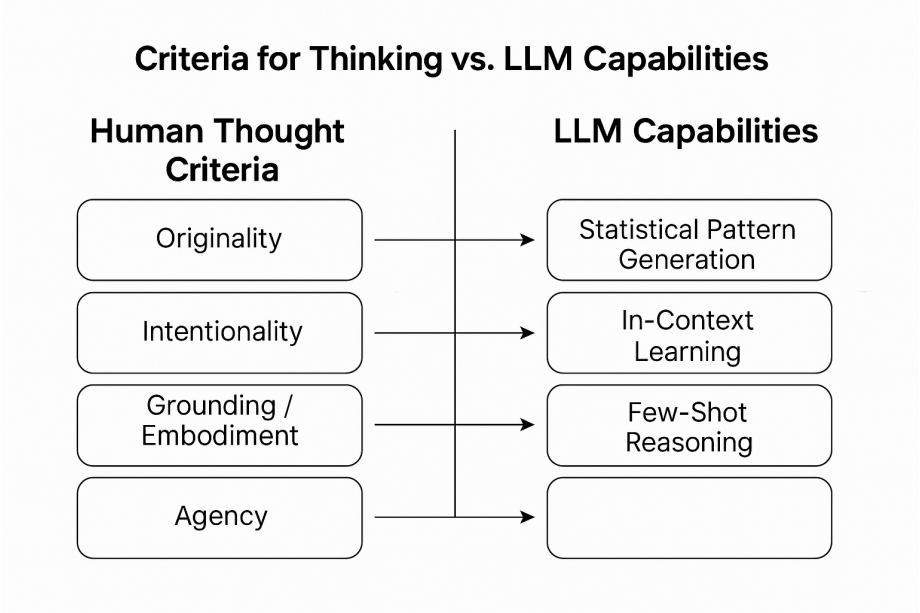

Figure 2: Mapping essential features of human thinking onto LLM strengths and gaps.

Considering these points, one can construct a tentative argument that **LLMs do exhibit a form of thinking**, albeit an alien one. They take in information (prompts) and process it to produce new information (responses) that often demonstrate insight, pattern recognition, and problem-solving. They use knowledge (stored in their weights from training) in a way that can be applied to novel situations, which parallels how humans use learned knowledge to reason about new questions. From a purely functional perspective, an LLM is an information-processing system that takes input, transforms internal representational states, and produces intelligent output. This fits a broad definition of thinking as information transformation to guide responses. Moreover, LLMs can be said to *generalise* beyond their training examples - they can handle questions they never saw answers to, by synthesising relevant pieces of training data. In that sense, they do something analogous to reasoning or inference.

One might also argue that the boundary between simulation and reality grows thin when the simulation is exceedingly good. If interacting with a future LLM gives the same experience as interacting with a knowledgeable, thoughtful person, at what point do we concede that the machine is thinking (at least in the conversational domain)? This touches on our intuitions and perhaps biases: we are inclined to attribute mind to anything that convincingly behaves like it has one (as evidenced by people who have felt that current chatbots were "sentient" simply because of their persuasive conversational skills). Adopting Dennett's idea of the *intentional stance*, we may treat the LLM as if it had beliefs and desires in order to predict its output, even if we know it doesn't actually have them[7]. In practical terms, if an AI consistently acts as if it understands and thinks, treating it as a thinking entity could be pragmatically useful.

On the other hand, a strong argument can be made that **LLMs lack critical elements of thinking**. They have no conscious awareness - there is nothing it "feels like" to be GPT-3, no inner qualitative experience accompanying its processing. If one views consciousness as essential to thought (as many would for genuinely understanding or creative thought), then LLMs are not thinkers, just tools. They also lack intentional states that are directed at the world: no beliefs about the world, no desires or goals. Any appearance of such states is a mirage created by the training data and prompt. The LLM does not *care* about truth or falsehood; it has no concept of truth, only what words might follow from other words. As a result, when it engages in reasoning, it is not undergirded by any commitment to logical consistency or reality - where a human reasoner has motivations (to get the right answer, to make sense, etc.), the LLM has none, and it sometimes free-associates into mistakes. One could say human thinking is guided by an understanding of the world and a purposeful structure (even subconscious purpose), whereas LLM processing is unguided beyond statistical fit to data. This fundamental difference might mean that what LLMs do is not thinking at all, but a sophisticated mimicry of the outputs of thinking beings.

In summary, whether LLMs should be considered thinkers hinges on which aspects of human thought we consider essential. If one emphasises behavioural and functional criteria - problem-solving ability, linguistic competence, flexibility in new tasks - then LLMs meet many of those benchmarks and arguably

engage in a form of thought (perhaps "thought without consciousness"). If one emphasises subjective understanding, intentionality, and genuine knowledge, then LLMs fall short, portraying only a hollow shell of thought. The next section will further articulate the skeptical viewpoint, reinforcing why many philosophers and scientists urge caution in saying that LLMs truly think.

5 Counterarguments and Philosophical Skepticism

Skeptics argue that however impressive LLMs are, calling what they do "thinking" is misleading if not outright incorrect. We now outline the major counterarguments to the claim that LLMs can think:

Lack of Understanding and Intentionality: The Chinese Room remains a potent analogy for LLMs. An LLM, like Searle's person in the room, shuffles symbols in sophisticated ways but does not *understand* the meanings of those symbols[4]. Every sentence produced by an LLM is generated without any comprehension; there is no mind behind the words thinking "this is true" or "this refers to that object." Searle's dictum that "syntax is not semantics" directly applies[4]. No matter how coherent the output, it is, from this perspective, meaningless to the machine. The machine isn't even *in* the semantic loop - it's we humans who imbue the output with meaning when we read it. Thus, to say the LLM is thinking could be seen as a category error: the machine is just executing code, while the thinking is happening in the user's interpretation. Additionally, since the LLM lacks any genuine beliefs or knowledge, it cannot be said to *think about* the world. It has been argued that true thought is always *about* something (this is the notion of intentionality) and involves mental representation of that thing. LLMs have representations (distributed vectors, activation patterns), but whether these count as "about" real-world entities is questionable given there's no intentional agent using them. Many would say these representations are more like a mirror of the training text than a model of reality. In short, skeptics maintain that without understanding, we don't have thinking - just verbal behaviour.

The Illusion of Originality: While LLMs can produce novel combinations of words, skeptics point out that this is not creativity in any meaningful sense. The model is essentially performing an auto-completion task; it has no insight, no spark of imagination, no problem it is trying to solve through creative thought. It does not set out to create or discover. Any appearance of creativity (say writing a poem in the style of Shakespeare on a modern topic) is because it has digested examples of Shakespearean style and modern topics and can statistically blend them. It's doing what mathematical mixture of patterns dictates, not experiencing an *aha* moment or making a leap. Furthermore, LLMs can also produce clichéd or incoherent results, revealing that they don't actually *know* when something is creative or banal. A human artist or thinker can judge and refine their creations with intent; an LLM cannot - it will continue to produce

whatever its training suggests, without an evaluative process (unless one is externally imposed via reinforcement learning, and even that is based on human judgments given to it). Thus, critics argue that LLMs are "blind" creators at best, and calling that true creativity dilutes the concept.

No Conscious Awareness (the Hard Problem): If one holds that thinking is intimately connected to consciousness - for instance, that the stream of thought is something experienced by a subject - then LLMs are definitively not thinkers. They are, at best, *zombies* that exhibit the outward signs of thinking with none of the inner life. Chalmers' hard problem highlights that we have no idea how or if an LLM could be conscious[6, 5, 16]. Indeed, Schneider argues that current AI systems fail every plausible test for phenomenal consciousness and thus cannot possess the kind of self-reflective awareness required for genuine deliberation[16]. There is nothing like a self–model or unified field of awareness in their architecture; an LLM is simply a large statistical associative memory with input–output mapping. Without consciousness, there is no genuine deliberation or reflection - the directed, self-aware activity that underpins human thinking. A person can think about their own thoughts (metacognition), notice mistakes in reasoning, and gain insight; an LLM cannot introspect or understand its states. It does not know that it knows (or that it does not know). Some philosophers (e.g. Dennett) might respond that consciousness is not a prerequisite for intelligent behaviour, yet even Dennett would concede that current AIs lack the *global workspace* or narrative centre that human minds possess. Therefore, labelling them "thinkers" risks anthropomorphism.

Dependence on Human Data and Lack of Autonomy: Another angle of skepticism is that LLMs are fundamentally derivative. They learned everything from human writings. They have no capacity to go beyond that source material except by recombination. They also do not formulate goals or questions on their own - they need a human prompt to do anything. In contrast, human thinkers are autonomous to a degree: we wonder, we question, we initiate conversations and inquiries driven by curiosity or purpose. An LLM left running with no prompt will just sit idle; it has no internal drive or agency. It doesn't *care* about truth, beauty, survival, or any of the motivations that underlie human thought processes. This lack of agency could be framed as lack of genuine thought: one might argue that thinking is an active process, something minds *do* to achieve aims or respond to stimuli. LLMs only react in a passive, pre-determined way to input. Their impressive outputs are ultimately a reflection of the data we gave them. Emily Bender and colleagues warn that these models are like convincing imitators without any grounding, and thus calling them intelligent or thinking is a dangerous misconception[13]. Gary Marcus and others have also noted that LLMs often fail at reasoning tasks that require understanding context or factual consistency, revealing that their apparent thinking is brittle and superficial. These critiques reinforce the idea that LLMs might best be viewed as *tools* created by real thinkers (humans), rather than independent

thinkers themselves.

Ethical and Conceptual Caution: Some skeptics point out that ascribing thinking or mind to LLMs may lead to misplaced trust or misguided decisions. If we think of them as thinkers, we might start expecting them to have judgment or to be responsible for their outputs, which they are not. We might anthropomorphise them and forget their true nature as pattern matchers, which can be risky (for example, using them in high-stakes decisions without human oversight because we mistakenly believe they "understand" the nuances). Philosophically, conflating simulation with reality could stall deeper understanding of intelligence; if we accept the simulation as the real thing too readily, we might miss the fact that something crucial is absent. For instance, if someone claims GPT-3 understands natural language, a skeptic will respond that this cheapens the notion of understanding and that we should reserve such terms for systems that can demonstrate comprehension in more than just superficial linguistic fluency.

Agency and Autonomy in LLMs: Most philosophical debates focus on whether LLMs *understand*; less attention is paid to whether they qualify as *agents*. Godfrey-Smith defines agency as goal-directed activity regulated by feedback from the environment [17]. On this view LLMs lack minimal agency: they possess no intrinsic goals, initiate no actions without an external prompt, and receive no sensorimotor feedback for error correction. Their outputs are consequents of a static objective - next-token prediction - rather than of self-generated aims. This deficit supports the claim that LLM cognition, if any, is *heteronomous*: entirely driven by user queries and training data. Acknowledging the agency gap clarifies why even functionally rich language behaviour may fall short of full-blooded thinking.

In sum, the counterarguments assert that LLMs are **merely impressive mimics** of thinking. They are products of human ingenuity that reproduce the outputs of thinking (language, answers, etc.) without the internal process that we consider true thought. The skepticism is not to deny that LLMs are useful or even "intelligent" in an engineering sense, but to deny them the status of having a mind. Simulation, no matter how life-like, is not equivalent to instantiating a mind, say the critics, echoing Searle[4]. Therefore, we should not conflate an LLM's performance with genuine understanding or reasoning; to do so would be to fall for an illusion generated by complex statistics.

6 Conclusion

The question of whether LLMs can think does not admit a simple yes or no answer - it forces us to examine what we mean by "think" and to confront the gap between appearance and reality in AI behaviour. On one level, LLMs undeniably *do* something cognitively interesting: they take in information and produce new, often insightful outputs. They can solve problems, carry on conversations, and mimic creative processes in writing. In a functional sense, they

perform many tasks we associate with thinking, and they do so using learned representations of the world encoded in language. If we adopt a functionalist or behaviourist framework, one could argue that LLMs engage in a form of thinking. After all, if an entity takes input, processes it in a way that reflects knowledge, and produces appropriate output, calling that process "thinking" might be a matter of pragmatic convenience (as Turing suggested when he replaced the question with the Imitation Game[1]). Furthermore, as AI systems become more advanced, the line between simulation and genuine cognitive ability may blur; we may reach a point where denying the term "thinking" to such systems feels like pure semantics or an insistence on an unobservable property.

However, as this paper has explored, there are compelling reasons to maintain a cautious distinction. Philosophically, thinking is not just about external performance but about internal states: having intentionality, understanding, and (for some theorists) consciousness. By those measures, current LLMs are deficient. They are *simulating mind* without *actually having mind* - hence our title, "Thinking Without Knowing," which captures the idea that an LLM can go through the *motions* of thinking (logical inference, language production) without any *knowledge* in the sense of justified true belief or conscious awareness. It thinks, but it does not know that it thinks, nor does it know in the rich way that humans know. Its “knowledge” is embedded in billions of weighted connections, accessible only through pattern complementation, not through explicit recall or understanding.

From a certain perspective, one might say LLMs are **as-if thinkers**: they behave *as if* they were thinking, and for many practical purposes, treating them as thinking entities (at least computationally) yields useful results. But whether they **are thinkers** in an ontological sense remains doubtful. This is not merely a pedantic point; it matters for how we design, use, and trust these systems. If we recognise that LLMs lack true understanding, we will be more alert to their failure modes (like producing plausible-sounding but incorrect answers) and more cautious about deploying them autonomously. If, conversely, one were to declare "LLMs can think now just like humans," one might overestimate their capabilities or even consider ethical questions about AI rights prematurely.

Our analysis suggests a middle ground: LLMs exhibit *proto-cognitive* abilities generated by an entirely different mechanism than human thought. They are powerful simulators of certain aspects of mind - especially linguistic expression and pattern recognition. In many ways, they are an extension of human cognitive artifact: trained on our collective writings, they reflect our knowledge and biases back to us. They do not *initiate* thought; they respond based on what they’ve seen. Thus, they might be best viewed as **tools that amplify human thought** rather than independent thinkers. They can assist with writing, brainstorming, and problem-solving, but always under the guidance (and guardrails) of human intellect that provides the true understanding of the context and the final verification of truth.

In conclusion, whether one calls LLMs "thinkers" is in part a semantic decision, but it has philosophical weight. This paper has argued that, strictly speaking, LLMs simulate thinking without possessing the full hallmarks of men-

tal life. They challenge our intuitions and theories about mind, much as a flying machine challenges our concept of flight: is a plane *flying* like a bird or merely achieving the same outcome by other means? In AI, the analogous question is: is the LLM *thinking* or just producing the outputs we associate with thinking by other means? Our exploration indicates that for now, it is the latter. LLMs are a mirror of mind, not a mind in their own right. They force us to refine our understanding of intelligence and remind us that genuine understanding might require more than just data and computation - it might require an experiencing subject, a grounded perspective, and self-driven agency, none of which current LLMs possess.

Ultimately, the exercise of asking if LLMs think leads us to deeper insight into both human cognition and the nature of these artificial models. It underscores the distinction between *performance* and *presence*: excellent performance in an intellectual task does not guarantee the presence of a mind. For the foreseeable future, LLMs should be respected as powerful *simulations* of thought, and used as such, but not confused with the real thing. In grappling with this distinction, we not only better understand LLMs, but also clarify what we consider most essential about thinking itself.

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